THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 05

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. Which of the following points lies on neither axis?

(a) (−4, 0) (b) (3, −2) (c) (0, 5) (d) (0, 0)

Ans: (b) A point lies on an axis if one coordinate is 0. Only (3, −2) has both coordinates non-zero.

2. The distance between the points (−2, 0) and (2, 0) is:

(a) 2 units (b) 4 units (c) 0 units (d) 8 units

Ans: (b) Both points lie on the x-axis: $|2 - (-2)| = 4$.

3. If the point P(a, 3) is at a distance of 5 units from the point (0, −1), then the value of a is:

(a) ±4 (b) 3 only (c) ±5 (d) ±3

Ans: (d) $a^2 + (3 + 1)^2 = 25 \Rightarrow a^2 = 9 \Rightarrow a = \pm 3$.

4. If $p(x) = 4x^2 - 3x + 7$, then p(0) is:

(a) 0 (b) 8 (c) 7 (d) −3

Ans: (c) $p(0) = 7$.

5. The degree of the polynomial $(x + 1)(x - 1)$ is:

(a) 0 (b) 1 (c) 2 (d) 3

Ans: (c) $(x + 1)(x - 1) = x^2 - 1$, degree 2.

6. The area of a rectangular garden is $(x^2 - 9)$ m² and its breadth is $(x - 3)$ m. Its length is:

(a) $(x^2 - 6)$ m (b) $(x - 3)$ m (c) $(x + 3)$ m (d) $(x + 9)$ m

Ans: (c) $x^2 - 9 = (x - 3)(x + 3) \Rightarrow$ length = $(x + 3)$ m.

7. Which of the following has a non-terminating repeating decimal expansion?

(a) $\frac{7}{25}$ (b) $\frac{5}{6}$ (c) $\frac{9}{40}$ (d) $\frac{3}{16}$

Ans: (b) $6 = 2 \times 3$ has the factor 3 (not 2 or 5) ⇒ non-terminating. The others have denominators 2^4, 5^2, $2^3 \times 5$.

8. The irrational number $\sqrt{3}$ lies between:

(a) 1 and 2 (b) 2 and 3 (c) 3 and 4 (d) 0 and 1

Ans: (a) $1^2 = 1 < 3 < 4 = 2^2$.

9. The value of $|3 - 7| + |-2|$ is:

(a) −6 (b) −2 (c) 6 (d) 2

Ans: (c) $|-4| + 2 = 4 + 2 = 6$.

10. The expression $x^2 - 6x + 9$ can be written as:

(a) $(x + 3)^2$ (b) $(x - 9)^2$ (c) $(x - 3)(x + 3)$ (d) $(x - 3)^2$

Ans: (d) $a^2 - 2ab + b^2$ with $a = x$, $b = 3$.

11. If $a + b + c = 10$ and $ab + bc + ca = 31$, then $a^2 + b^2 + c^2$ is:

(a) 100 (b) 38 (c) 69 (d) 41

> **Ans: (b)** $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = 100 - 62 = 38$.

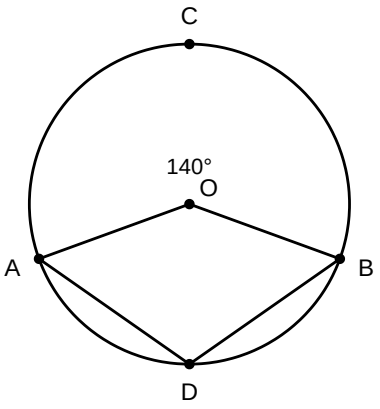
12. In a circle with centre O, chords AB and CD are equal. If the perpendicular distance of AB from O is 7 cm, the perpendicular distance of CD from O is:

(a) cannot be found (b) 3.5 cm

(c) 7 cm (d) 14 cm

> **Ans: (c)** Equal chords of a circle are equidistant from the centre.

13. In the figure, O is the centre of the circle and $\angle AOB = 140°$. If D is a point on the minor arc AB, then $\angle ADB$ is:



(a) 220° (b) 110° (c) 70° (d) 140°

> **Ans: (b)** Angle on the major arc = 70°; ADCB is cyclic ⇒ $\angle ADB = 180° - 70° = 110°$.

14. The area of an equilateral triangle of side 8 cm is:

(a) $64\sqrt{3}$ cm² (b) $8\sqrt{3}$ cm² (c) $16\sqrt{3}$ cm² (d) $32\sqrt{3}$ cm²

> **Ans: (c)** $(\sqrt{3}/4) \times 8^2 = 16\sqrt{3}$.

15. The area of a triangle with sides 9 cm, 10 cm and 17 cm is:

(a) 36 cm² (b) 45 cm² (c) 72 cm² (d) 90 cm²

> **Ans: (a)** $s = 18$; $\sqrt{(18 \times 9 \times 8 \times 1)} = \sqrt{1296} = 36$.

16. Two dice are thrown together. The probability that the sum of the numbers is 12 is:

(a) $\frac{1}{6}$ (b) $\frac{1}{36}$ (c) $\frac{1}{18}$ (d) $\frac{1}{12}$

> **Ans: (b)** Only (6, 6) out of 36 outcomes.

17. A letter is chosen at random from the word STATISTICS. The probability that it is T is:

(a) $\frac{2}{5}$ (b) $\frac{3}{10}$ (c) $\frac{1}{5}$ (d) $\frac{1}{10}$

> **Ans: (b)** T appears 3 times in 10 letters.

18. If the n-th term of a sequence is $t_n = n^2 + 1$, then t_9 is:

(a) 19 (b) 10 (c) 81 (d) 82

> **Ans: (d)** 81 + 1 = 82.

19. **Assertion (A):** The point (−3, 0) lies on the x-axis.
Reason (R): Every point on the x-axis has ordinate 0.

> **Ans: (a)** Ordinate of (−3, 0) is 0, so it lies on the x-axis; R explains A.

20. **Assertion (A):** $(a + b)^2 = a^2 + b^2$ for all real numbers a and b.
Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

> **Ans: (d)** R is the true identity; the extra term 2ab makes A false in general (e.g. a = b = 1: $4 \neq 2$).

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A courier company charges ₹40 for the first 500 g and ₹15 for every additional 100 g (or part thereof). Write the charge for n additional units of 100 g and find the charge for a parcel of 1.2 kg.

> Charge = 40 + 15n (1). 1.2 kg = 1200 g; additional 700 g = 7 units ⇒ 40 + 105 = **₹145** (1)

22. Without performing division, show that $\frac{49}{2^3 \times 5^4 \times 7}$ is a terminating decimal and state the number of decimal places.

> Simplify: $49/(2^3 \times 5^4 \times 7) = 7/(2^3 \times 5^4)$ (½). Only 2 and 5 in the denominator ⇒ terminating (½). Decimal places = greater power = **4** (1).

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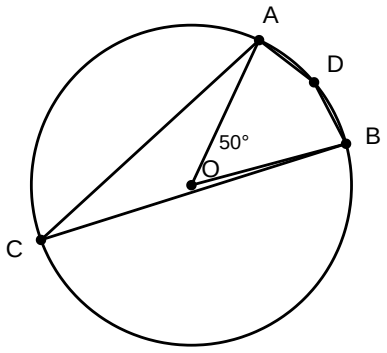
Find two rational numbers between $\frac{5}{8}$ and $\frac{3}{4}$.

> 5/8 = 10/16 = 20/32 and 3/4 = 24/32; so e.g. 11/16 and 23/32 (1 each). Check: 5/8 < 11/16 < 23/32 < 3/4.

23. Factorise: $49x^2 + 84xy + 36y^2$.

> $= (7x)^2 + 2(7x)(6y) + (6y)^2$ (1) = **$(7x + 6y)^2$** (1)

24. In the figure, O is the centre of the circle and ∠AOB = 50°. Find ∠ACB and ∠ADB.



> ∠ACB = ½ × 50° = **25°** (1)
> ACBD is cyclic ⇒ ∠ADB = 180° − 25° = **155°** (1)

25. Find the second, third and fourth terms of the sequence defined by $t_1 = 3$ and $t_n = 2t_{n-1} + n$ for $n \geq 2$.

> $t_2 = 2(3) + 2 = 8$; $t_3 = 2(8) + 3 = 19$; $t_4 = 2(19) + 4 = 42$. (½ for method + 1½ for values)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 0.363636… + 0.272727… as a fraction in the form p/q.

> 0.3636… = 36/99 = 4/11 (1); 0.2727… = 27/99 = 3/11 (1)
> Sum = 7/11 (1)

27. A coin was tossed 80 times and the number of heads was recorded:

Heads	3	2	1	0
Frequency	10	28	30	12

Find the probability of getting (i) at least 2 heads, (ii) exactly 1 head, (iii) no head.

(i) (10 + 28)/80 = **19/40** (1) (ii) 30/80 = **3/8** (1) (iii) 12/80 = **3/20** (1)

OR

A die was thrown 60 times with the following results:

Outcome	1	2	3	4	5	6
Frequency	8	12	10	9	11	10

Find the probability of getting (i) an even number, (ii) a number greater than 4, (iii) the number 1.

(i) (12 + 9 + 10)/60 = **31/60** (1) (ii) (11 + 10)/60 = **7/20** (1) (iii) 8/60 = **2/15** (1)

28. Find the distance between the points A(−6, 7) and B(−1, −5), and the coordinates of the midpoint of AB.

AB = √[(−1 + 6)² + (−5 − 7)²] (1) = √(25 + 144) = **13 units** (1)
Midpoint = ((−6 − 1)/2, (7 − 5)/2) = **(−7/2, 1)** (1)

29. Factorise $x^3 - 6x^2 + 12x - 8$ and hence find its value for x = 12.

$= x^3 - 3 \cdot x^2 \cdot 2 + 3 \cdot x \cdot 2^2 - 2^3 = (x - 2)^3$ (2)
For x = 12: 10^3 = **1000** (1)

30. Prove that the perpendicular drawn from the centre of a circle to a chord bisects the chord.

Given: chord AB of a circle with centre O; OM ⊥ AB. To prove: AM = MB. Join OA and OB. (½)
In △OMA and △OMB: ∠OMA = ∠OMB = 90°, OA = OB (radii), OM = OM (common) (1½)
So △OMA ≅ △OMB (RHS) (½) ⇒ AM = MB (CPCT). (½)

31. The sides of a triangular park are in the ratio 7 : 24 : 25 and its perimeter is 168 m. Find its area using Heron's formula.

Sides = 7k, 24k, 25k; 56k = 168 ⇒ k = 3: 21 m, 72 m, 75 m (1)
s = 84; area = √(84 × 63 × 12 × 9) (1) = **756 m²** (1)

OR

Find the area of a triangle with sides 9 cm, 40 cm and 41 cm using Heron's formula.

s = 45 (1); area = √(45 × 36 × 5 × 4) (1) = √32400 = **180 cm²** (1)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

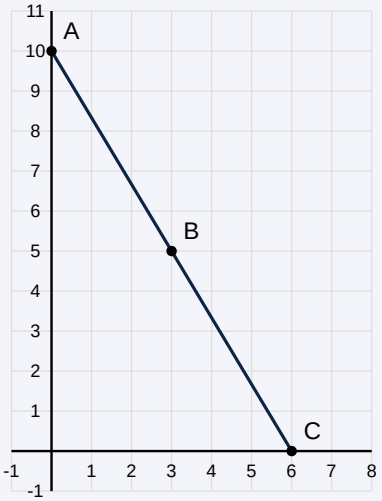
Internal choice is provided in two questions.

32. Riya buys x pencils at ₹5 each and y erasers at ₹3 each and spends exactly ₹30.
(a) Write this as a linear equation in x and y, in the form ax + by + c = 0, and write a, b, c. (1 mark)
(b) Complete a table of three solutions (x, y). (1 mark)
(c) Draw the graph of the equation on a graph paper. (2 marks)
(d) Use the graph to find the number of erasers she buys if she buys 3 pencils. (1 mark)

(a) 5x + 3y = 30 ⇒ 5x + 3y − 30 = 0; a = 5, b = 3, c = −30 (1)
(b)

x	0	3	6
y	10	5	0

(1)



(c) Plot (0, 10), (3, 5), (6, 0) (1) and join with a straight line (1): (d) At x = 3, y = **5 erasers** (1)

33. (a) How many odd numbers lie between 100 and 400? (b) Find their sum. (c) Is 305 a term of this list? Give a reason. (2 + 2 + 1 marks)

Numbers: 101, 103, …, 399; AP with a = 101, d = 2, l = 399 (1)
(a) n = (399 − 101)/2 + 1 = **150** (1)
(b) S = 150/2 × (101 + 399) = 75 × 500 = **37500** (2)
(c) Yes, 305 is odd and lies between 100 and 400 (1)

34. The lifetimes of 1000 electric bulbs were recorded:

Lifetime (hours)	0 – 300	300 – 600	600 – 900	900 – 1200
Number of bulbs	40	160	450	350

A bulb is chosen at random. Find the probability that its lifetime is (i) less than 300 hours, (ii) between 300 and 900 hours, (iii) more than 900 hours.

Total = 1000 (1)
(i) 40/1000 = **1/25** (1½)
(ii) (160 + 450)/1000 = 610/1000 = **61/100** (2)
(iii) 350/1000 = **7/20** (½ + ½)

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A survey of 200 families recorded the number of children in each family:

Children	0	1	2	3
Families	22	90	70	18

A family is chosen at random. Find the probability that it has (i) exactly 2 children, (ii) at most 1 child, (iii) more than 2 children.

Total = 200 (1)
(i) 70/200 = **7/20** (1½)
(ii) (22 + 90)/200 = 112/200 = **14/25** (2)
(iii) 18/200 = **9/100** (½ + ½)

35. The length of a rectangular park is 3 m more than its breadth and its area is 130 m². Form a quadratic expression, factorise it to find the dimensions, and find the cost of fencing the park at ₹40 per metre.

Breadth = b, length = b + 3 ⇒ b(b + 3) = 130 ⇒ $b^2 + 3b - 130 = 0$ (1½)
$b^2 + 13b - 10b - 130 = (b + 13)(b - 10) = 0$ ⇒ b = 10 (1½)
Breadth = 10 m, length = 13 m (½); perimeter = 2(10 + 13) = 46 m (½); cost = 46 × 40 = **₹1840** (1)

OR

A mosaic is made of 2 big square tiles of side x, 9 long strips of size x × 1 and 4 small unit squares, so its area is $(2x^2 + 9x + 4)$. Factorise the expression to find the dimensions of the rectangle formed. Find its perimeter and area when x = 5.

$2x^2 + 9x + 4 = 2x^2 + 8x + x + 4$ (1½) $= 2x(x + 4) + 1(x + 4) =$ **$(2x + 1)(x + 4)$** (1½)
Dimensions: $(2x + 1)$ and $(x + 4)$ (½). For x = 5: 11 and 9 units ⇒ perimeter = 2(11 + 9) = **40 units** (1), area = 99 square units (½)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A triangular plot of land has sides 25 m, 29 m and 36 m.

(i) Find the semi-perimeter of the plot.

s = (25 + 29 + 36)/2 = **45 m**

(ii) Find the area of the plot using Heron's formula.

√(45 × 20 × 16 × 9) = √129600 = **360 m²**

(iii) The plot is to be levelled at ₹5 per m². Find the total cost.

Cost = 360 × 5 (1) = **₹1800** (1)

OR

(iii) Find the altitude of the plot on its longest side.

½ × 36 × h = 360 (1) ⇒ h = **20 m** (1)

37. In a school sports day, 120 entries were received for four events:

Event	Races	Jumps	Throws	Relays
Entries	36	30	24	30

One entry form is picked at random.

(i) Find the probability that it is for a race.

36/120 = **3/10**

(ii) Find the probability that it is not for a throw.

96/120 = **4/5**

(iii) Find the probability that it is for a jump or a relay.

(30 + 30)/120 (1) = **1/2** (1)

OR

(iii) If 20 more entries are received, all for throws, find the new probability that a randomly picked entry is for a throw.

Throws = 44, total = 140 (1) ⇒ 44/140 = **11/35** (1)

38. Kabir saves money in a jar. He starts with ₹200 and adds ₹40 every week. The amount in the jar after w weeks is M = 200 + 40w.

(i) How much money is there in the jar after 6 weeks?

M = 200 + 240 = **₹440**

(ii) What does the number 40 in the equation represent?

The amount (₹) added to the jar every week.

(iii) After how many weeks will the jar have ₹800?

> 200 + 40w = 800 (1) ⇒ w = **15 weeks** (1)

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(iii) Find the amount after 12 weeks and its increase over the amount after 4 weeks.

> M(12) = **₹680**; M(4) = 360 (1); increase = **₹320** (1)